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Compare Permissions on Regular Files and Directories

Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.

Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

```
asus@LAPTOP-DFL0EA7B: ~/l × + v
asus@LAPTOP-DFL0EA7B:~$ mkdir ~/linux_assignment
mkdir: /home/asus/linux_assignment: File exists
asus@LAPTOP-DFL0EA7B:~$ cd linux_assignment
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ touch sample.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls
sample.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ mkdir sample_dir
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls
sample.txt  sample_dir
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ touch sample_dir/test.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls
sample.txt  sample_dir
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ nano sample.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls -l
total 8
-rw-r--r-- 1 asus asus 53 Aug 20 18:12 sample.txt
drwxr-xr-x 2 asus asus 4096 Aug 20 18:11 sample_dir
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls -l sample_dir
total 0
-rw-r--r-- 1 asus asus 0 Aug 20 18:11 test.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ |
```

Permission Regular File (sample.txt) – Test Observation Directory (sample_dir) – Test Observation

1. 400 (r--) Try to read and modify the file Can read; cannot modify Try to list and enter the directory Can read; cannot modify

```
asus@LAPTOP-DFL0EA7B:~$ mkdir ~/linux_assignment
mkdir: /home/asus/linux_assignment: File exists
asus@LAPTOP-DFL0EA7B:~$ cd linux_assignment
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ touch sample.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls
sample.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ mkdir sample_dir
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls
sample.txt  sample_dir
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ touch sample_dir/test.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls
sample.txt  sample_dir
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ nano sample.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls -l
total 8
-rw-r--r-- 1 asus asus 53 Aug 20 18:12 sample.txt
drwxr-xr-x 2 asus asus 4096 Aug 20 18:11 sample_dir
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls -l sample_dir
total 0
-rw-r--r-- 1 asus asus 0 Aug 20 18:11 test.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 400 sample.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ ls -l sample.txt
-r----- 1 asus asus 53 Aug 20 18:12 sample.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ cat sample.txt
echo "This is a sample file for permission testing."
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ echo "Testing write permission" >> sample.txt
-bash: sample.txt: Permission denied
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ cat sample.txt
echo "This is a sample file for permission testing."
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 400 sample_dir
ls -ld sample_dir
ls sample_dir
cd sample_dir
dr----- 2 asus asus 4096 Aug 20 18:11 sample_dir
test.txt
-bash: cd: sample_dir: Permission denied
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ cd ..
asus@LAPTOP-DFL0EA7B:~$ |
```

2. 200 (-w-) Try to read and modify the file Cannot read; can modify/write Try to list, enter, and create a file Cannot normally list, enter, or create because x is missing.

```
asus@LAPTOP-DFL0EA7B: ~/linux_assignment$ chmod 200 sample.txt
cat sample.txt
echo "New text" >> sample.txt
cat: sample.txt: Permission denied
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 200 sample_dir
ls sample_dir
cd sample_dir
touch new.txt
ls: cannot open directory 'sample_dir': Permission denied
-bash: cd: sample_dir: Permission denied
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ |
```

3. 100 (--x) Try to read and execute the file Cannot read; execute permission alone does not make an ordinary text file executable. Try to list and enter the directory Cannot list contents but can traverse/enter it.

```
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 100 sample.txt
cat sample.txt
cat: sample.txt: Permission denied
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 100 sample_dir
ls sample_dir
cd sample_dir
ls: cannot open directory 'sample_dir': Permission denied
asus@LAPTOP-DFL0EA7B:~/linux_assignment/sample_dir$ |
```

4. 500 (r-x) Try to read and modify the file Can read, cannot modify. Try to list and enter the directory Can list and enter, but cannot create/delete files

```
asus@LAPTOP-DFL0EA7B:~/linux_assignment/sample_dir$ cd ..
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 500 sample.txt
cat sample.txt
echo "New text" >> sample.txt
echo "This is a sample file for permission testing."
New text
New text
-bash: sample.txt: Permission denied
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 500 sample_dir
ls sample_dir
cd sample_dir
test.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment/sample_dir$ |
```

5. 600 (rw-) Try to read and modify the file Can read and modify. Try to list, enter, and create a file Cannot properly access/traverse because execute permission is missing.

```
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 600 sample.txt
cat sample.txt
echo "Modified text" >> sample.txt
echo "This is a sample file for permission testing."
New text
New text
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 600 sample_dir
ls sample_dir
cd sample_dir
test.txt
-bash: cd: sample_dir: Permission denied
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ |
```

6. 700 (rwx) Try to read, modify, and execute the file Owner has read, write, and execute permissions. Try to list, enter, create, and delete files Owner can list, enter, create, and delete files.

```
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 700 sample.txt
ls -l sample.txt
cat sample.txt
echo "Modified again" >> sample.txt
-rwx----- 1 asus asus 85 Aug 20 18:38 sample.txt
echo "This is a sample file for permission testing."
New text
New text
Modified text
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ chmod 700 sample_dir
ls sample_dir
cd sample_dir
touch new.txt
ls
rm new.txt
cd ..
test.txt
new.txt test.txt
asus@LAPTOP-DFL0EA7B:~/linux_assignment$ |
```

Questions

1. What does read (r) permission allow you to do with a regular file?
➔ It allows the user to **read/view the contents of the file**.
2. What does read (r) permission allow you to do with a directory?
➔ It allows the user to **list the names of files and directories inside it**.
3. What does write (w) permission mean for a regular file?
➔ It allows the user to **modify the contents of the file**.
4. What does write (w) permission mean for a directory?
➔ It allows the user to **create, delete, and rename entries** in the directory, when the required execute permission is also available.
5. What does execute (x) permission mean for a regular file?
➔ It allows the file to be **executed as a program**, if the file is actually executable.
6. Why is execute (x) permission important for accessing a directory?
➔ For a directory, x means **traverse/search permission**. It allows the user to enter the directory and access its entries.

7. Which permission combination allows the owner to fully access a regular file?
➔ 700 (rwx) gives the owner read, write, and execute permission.
8. Which permission combination allows the owner to fully access a directory?
➔ 700 (rwx) gives the owner permission to list, enter/traverse, create, delete, and modify directory entries.
9. Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.
➔ Because a regular file contains **data**, while a directory contains **references to files and other directories**.
Regular file -> r = read contents, w = modify contents, x = execute
Directory -> r = list contents, w = create/delete/rename entries, x = enter/traverse/access